

Introduction to Machine Learning

What is Machine Learning?

Machine learning is a subfield of artificial intelligence that enables systems to learn and improve from experience without being explicitly programmed. It focuses on developing algorithms that can access data and use it to learn for themselves.

The primary goal of machine learning is to allow computers to learn automatically without human intervention and adjust actions accordingly. This is achieved by training algorithms on large datasets and allowing them to identify patterns and make decisions with minimal human input.

Supervised vs Unsupervised Learning

Supervised learning is a type of machine learning where the model is trained on labeled data. The algorithm learns to map input features to output labels by studying examples where the correct answer is known. Common supervised learning tasks include classification and regression.

Unsupervised learning, in contrast, works with unlabeled data. The algorithm must find hidden structure or patterns in the input data on its own. Clustering, dimensionality reduction, and anomaly detection are common unsupervised learning tasks.

Semi-supervised learning sits between these two paradigms, using a small amount of labeled data combined with a larger amount of unlabeled data during training. This is particularly useful when labeling data is expensive or time-consuming.

Overfitting and Regularization

Overfitting occurs when a machine learning model learns the training data too well, capturing noise and random fluctuations as if they were real patterns. An overfitted model performs very well on training data but fails to generalize to new, unseen examples.

Regularization is a set of techniques used to prevent overfitting by adding a penalty term to the loss function that discourages the model from learning overly complex patterns. Common regularization techniques include L1 regularization (Lasso), L2 regularization (Ridge), and dropout for neural networks.

Cross-validation is another important technique for detecting and preventing overfitting. By training on multiple subsets of data and evaluating on held-out sets, we can estimate how well the model will generalize to new data.

Neural Networks and Deep Learning

Neural networks are computational models inspired by the structure and function of the human brain. They consist of layers of interconnected nodes (neurons) that process information and learn to represent complex patterns through training.

Deep learning refers to neural networks with many layers (deep architectures). These models have achieved remarkable success in tasks such as image recognition, natural language processing, and speech recognition by learning hierarchical representations of data.

Backpropagation is the core algorithm used to train neural networks. It computes the gradient of the loss function with respect to each weight in the network, allowing the model to update its parameters in the direction that reduces prediction error.

Model Evaluation Metrics

Accuracy is the most straightforward metric, measuring the proportion of correctly classified instances. However, accuracy can be misleading when classes are imbalanced — a model that always predicts the majority class can achieve high accuracy while being useless.

Precision measures what fraction of positive predictions were actually correct. Recall (sensitivity) measures what fraction of actual positives were correctly identified. The F1 score is the harmonic mean of precision and recall, providing a balanced measure.

For regression tasks, common metrics include Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and Mean Absolute Error (MAE). The choice of metric should reflect the actual costs of different types of errors in the application domain.